

SegmentGDA: Training-Free Few-Shot Segmentation via Gaussian Discriminant Analysis on Vision Foundation Features

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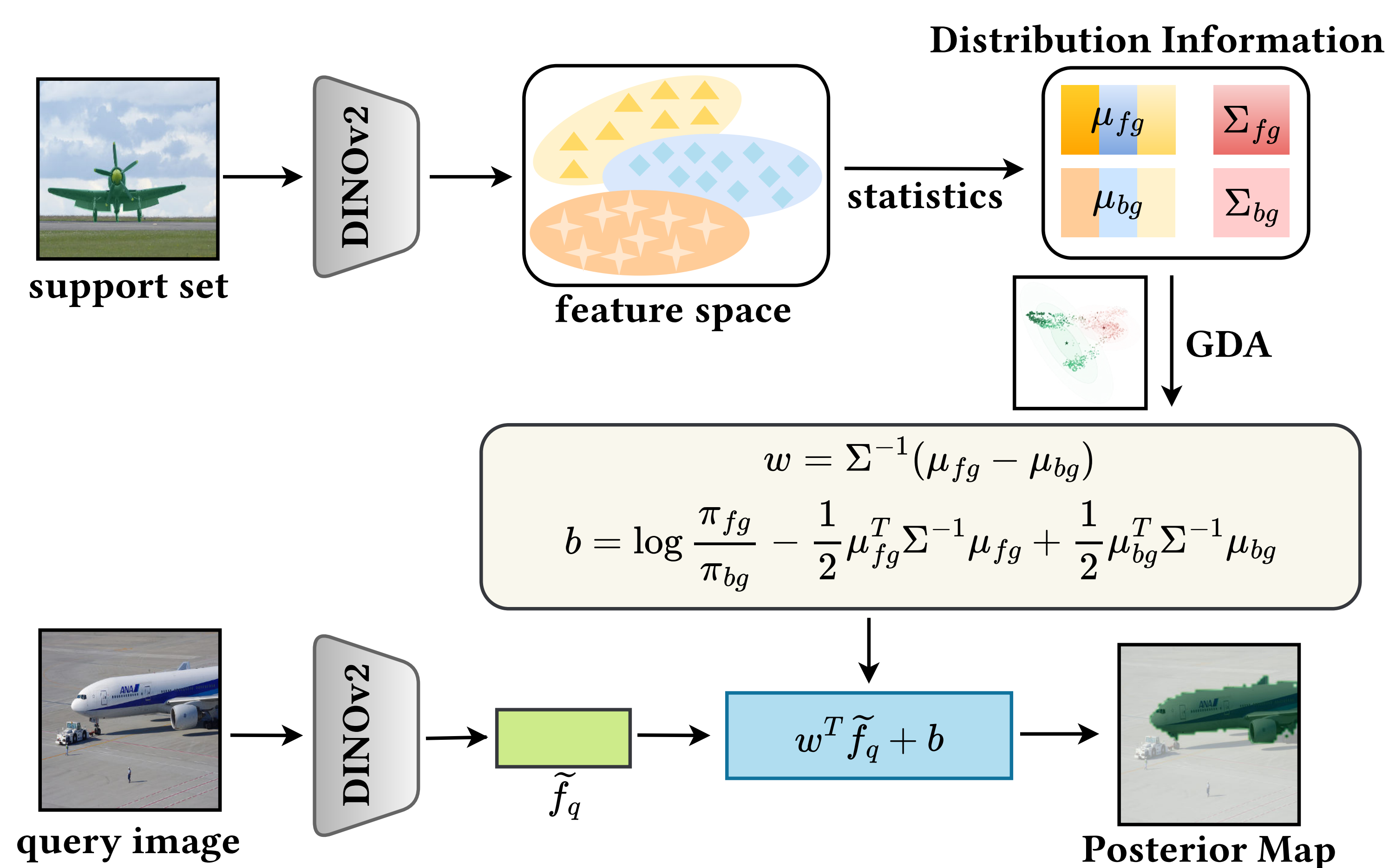
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01 Motivation

Segment a novel object from only a few annotated support examples.

Local patch matching can miss object parts and select background clutter.

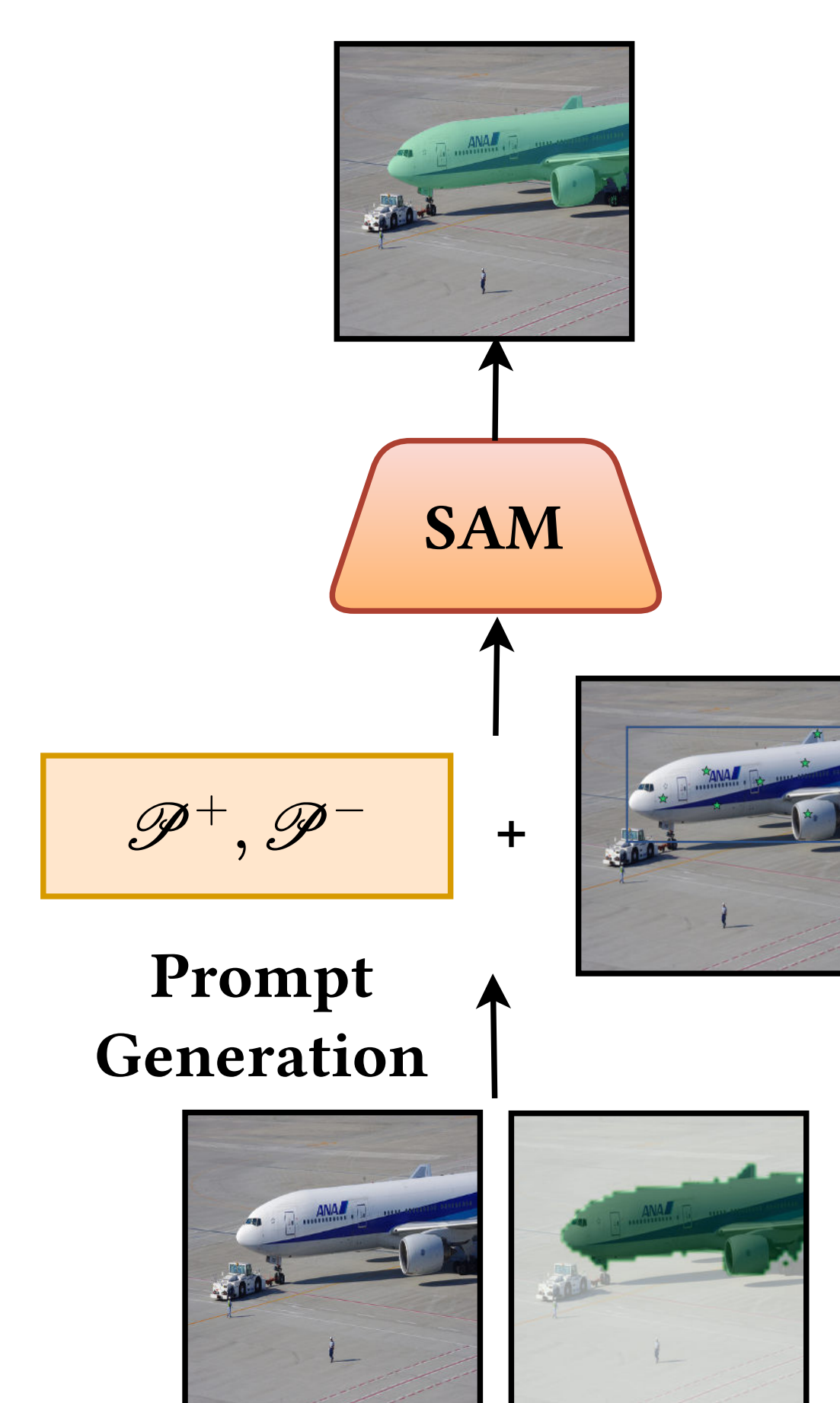
03 Method overview



02 Core idea

Model foreground and background as distributions in frozen foundation features.

Bayesian posteriors guide SAM prompts. No task-specific training or fine-tuning.



1 Frozen features

DINOv2 ViT-L/14 features, signed square-root transform and L2 normalization.

2 Gaussian inference

Two class means and a shared covariance yield patch-level foreground posteriors.

3 SAM refinement

Diverse positive and negative points guide frozen SAM ViT-H to iteratively correct errors.

04 Benchmark results

mIoU (%) Gain = 5-shot minus 1-shot

Dataset	1-shot	5-shot	Gain (pp)
COCO-20 ⁱ	42.1	62.5	+20.4
FSS-1000	82.9	89.8	+6.9
LVIS-92 ⁱ	26.2	41.8	+15.6
PASCAL-5 ⁱ	59.5	82.2	+22.7

PASCAL-5ⁱ, 5-shot: 82.2 vs. Matcher 74.0 and GF-SAM 82.6.

More support examples improve the estimates. One-shot performance remains a limitation.

05 Ablation

COCO-20ⁱ, 5-shot mIoU (%)

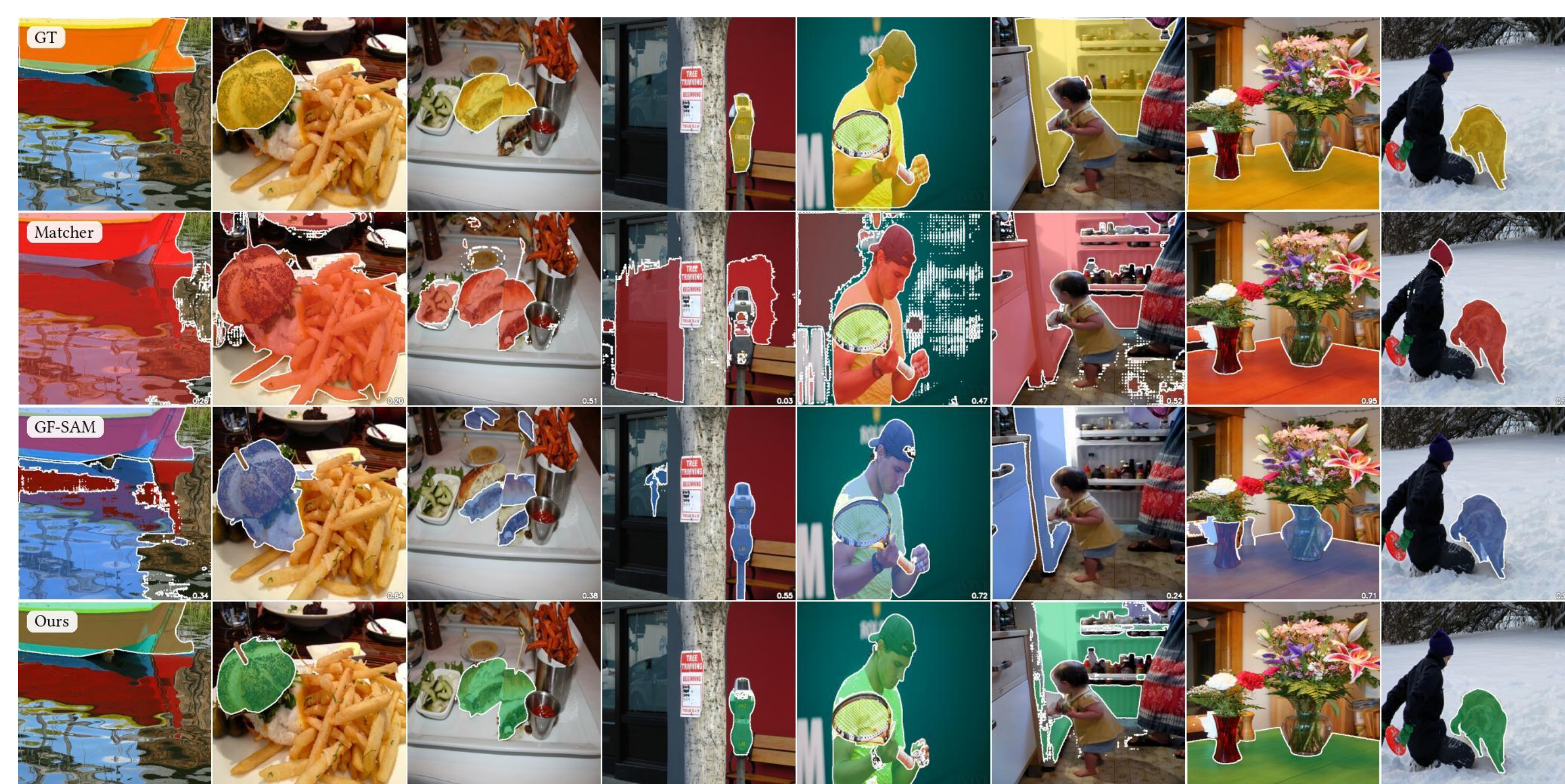
Point-wise matching	53.6
Mean-only Euclidean	59.9
Shared diagonal covariance	60.8
Shared full covariance	62.5

+8.9 pp

over point-wise matching in the controlled ablation.

06 Qualitative comparisons

COCO-20ⁱ



Rows: ground truth / Matcher / GF-SAM / SegmentGDA. Selected 5-shot episodes.

FSS-1000



TAKEAWAY Global statistics improve multi-shot mask transfer.

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CODE